

brocc_mediation

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libraries and directories

```
library(dplyr)
library(tibble)
library(tidyr)
library(ggplot2)
library(mediation)
library(ggfortify)
library(ggpubr)
library(tweedie)
library(statmod)
library(AICcmodavg)
library(stringr)
set.seed(3)
```

```
setwd("/Users/alexieva/Documents/Projects/Analysis/broccoli project/02_pubAnalysis/Prediction1_mediation")
load("/Users/alexieva/Documents/Projects/Analysis/broccoli project/02_pubAnalysis/Prediction1_mediation.Rsave")
```

Input data

```
## microbiome asv table; this is at time 0 as per previous exploratory data analysis
# only taxa that were previously identified as correlated with SFN and SFN-NIT, whose taxa names are in SFN_taxa
SFN_taxa <- read.csv("/Users/alexieva/Documents/Projects/Analysis/broccoli project/01_explAnalysis/input/SFN_taxa.csv")

MB_asvtab <- read.table("/Users/alexieva/Documents/Projects/Analysis/broccoli project/01_explAnalysis/input/MB_asvtab.csv",
  dplyr::select(SFN_taxa$x) %>%
  rownames_to_column("subject_id")

## SFN data
# using cumulative SFN, SFN-NIT, and SFN tot across all time points
# note that there are a few alfalfa eaters here that get filtered out in the process of making combined
Urine_SFN <- read.csv("/Users/alexieva/Documents/Projects/Analysis/broccoli project/00_data/SFN_Data/Urine_SFN.csv",
  header = TRUE) %>%
  group_by(subject_id) %>%
  summarise(sum_SFN = sum(SFN),
    sum_SFN_Cys = sum(SFN_Cys),
    sum_SFN_NAC = sum(SFN_NAC),
```

```

sum_SFN_CG = sum(SFN_CG),
sum_SFN_GSH = sum(SFN_GSH),
sum_SFN_NIT = sum(SFN_NIT),
sum_SFN_Tot = sum(SFN_Tot))
# %>% spread(key = "SFN_metabolites", value = "quantity")

## diet data
# make a list of the "significant" diet factors we are interested in
diet_compnts <- c("diet_Chol_mg", "diet_SatFat_g", "diet_Alc_g", "diet_TotSolFib_g",
                  "diet_MonSac_g", "diet_Disacc_g", "diet_BetaCaro_mcg", "diet_Vit_B1_mg",
                  "diet_Vit_B6_mg", "diet_Fol.DFE_mcg_DFE", "diet_Mang_mg", "diet_Copp_mg",
                  "diet_TotFib_g", "diet_OCarb_g", "diet_Iron_mg", "diet_TotInsolFib_g")
diet <- read.csv("/Users/alexieva/Documents/Projects/Analysis/broccoli project/00_data/Diet/New_Diet_Sm
dplyr::select(c("subject_id", all_of(diet_compnts)))

# write.table(diet, "../input_data/dietData_clean.txt", row.names = F, quote = F)

# match data based on subject id
med_input <- MB_asvtab %>%
  inner_join(Urine_SFN, by = "subject_id") %>%
  inner_join(diet, by = "subject_id")

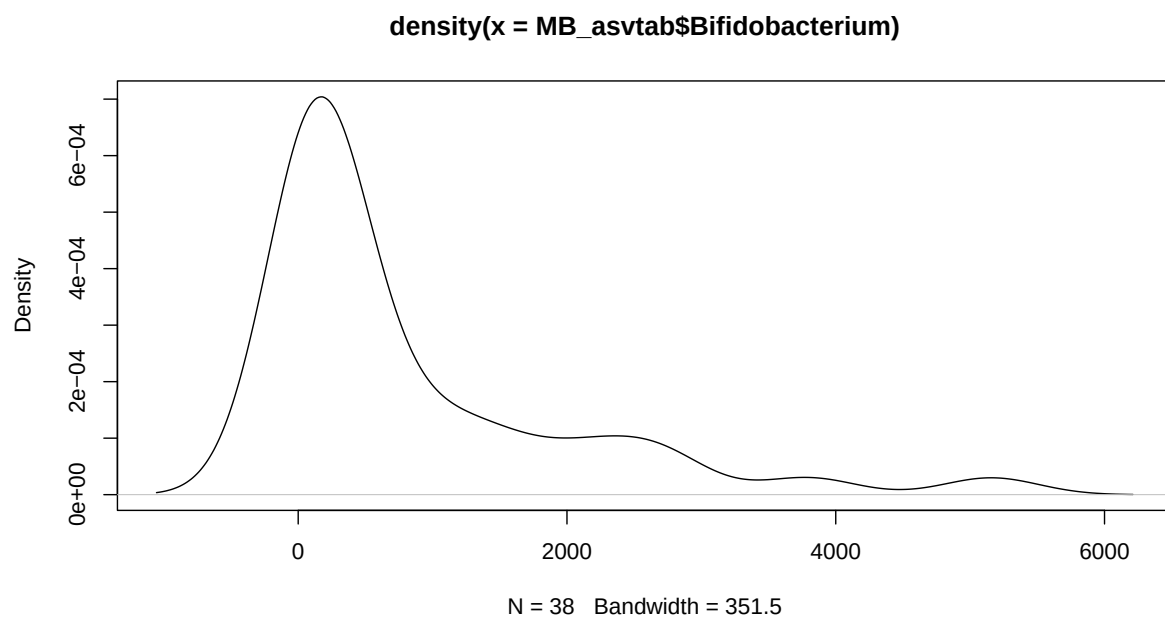
# write.table(med_input, "../input_data/combinedData_clean.txt", row.names = F, quote = F)

```

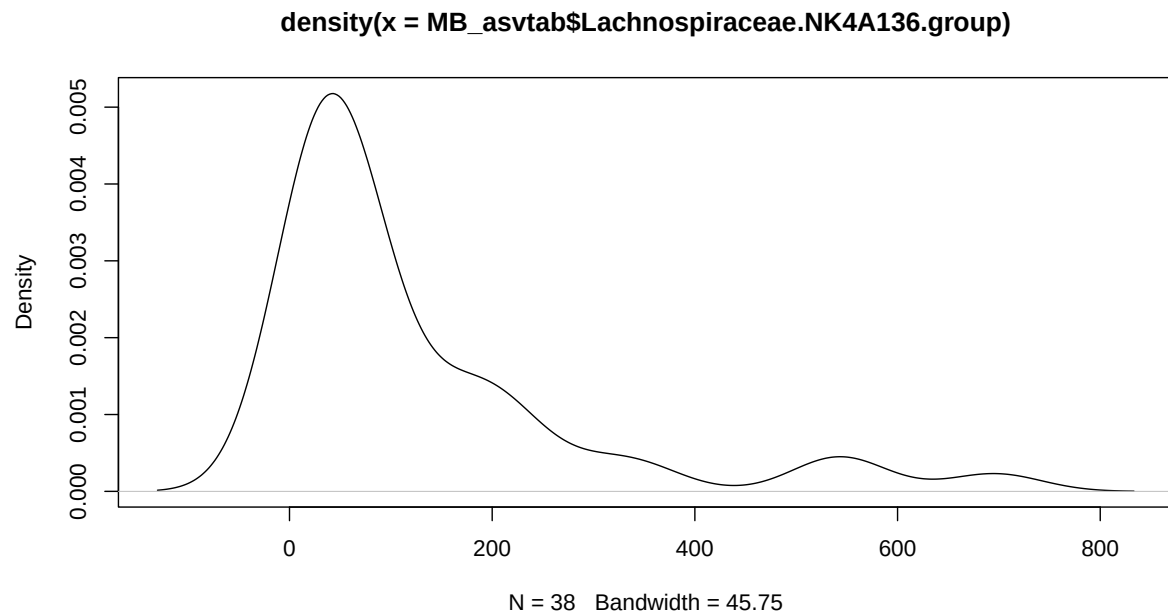
LEFT OFF: rerunning this below with insol fiber data frames

check distributions of variables

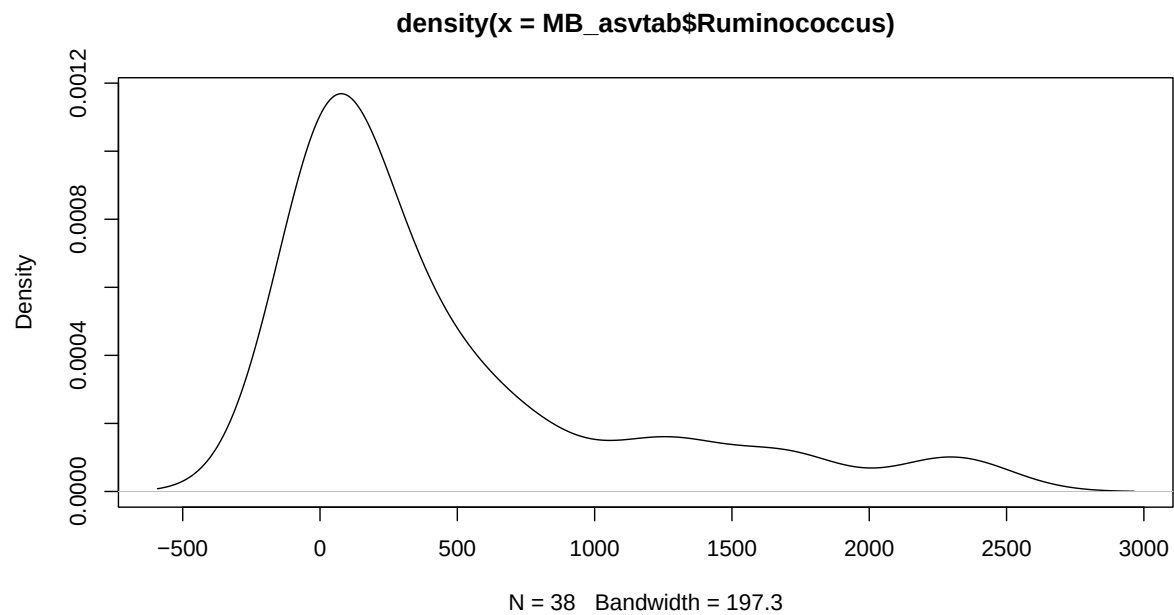
```
plot(density(MB_asvtab$Bifidobacterium))
```



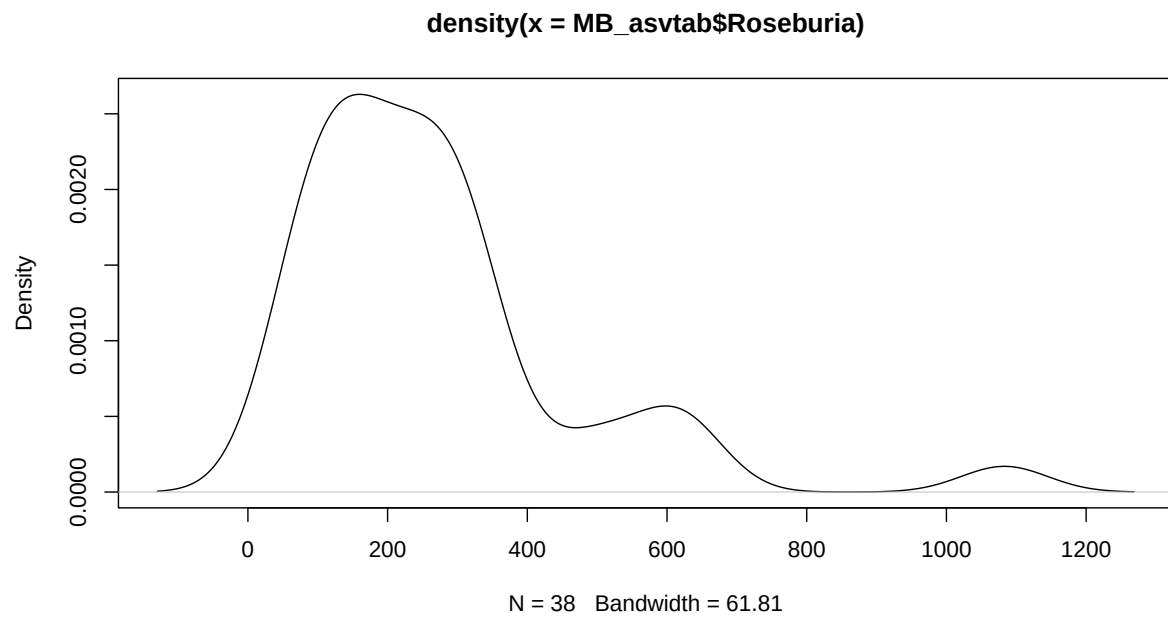
```
plot(density(MB_asvtab$Lachnospiraceae.NK4A136.group))
```



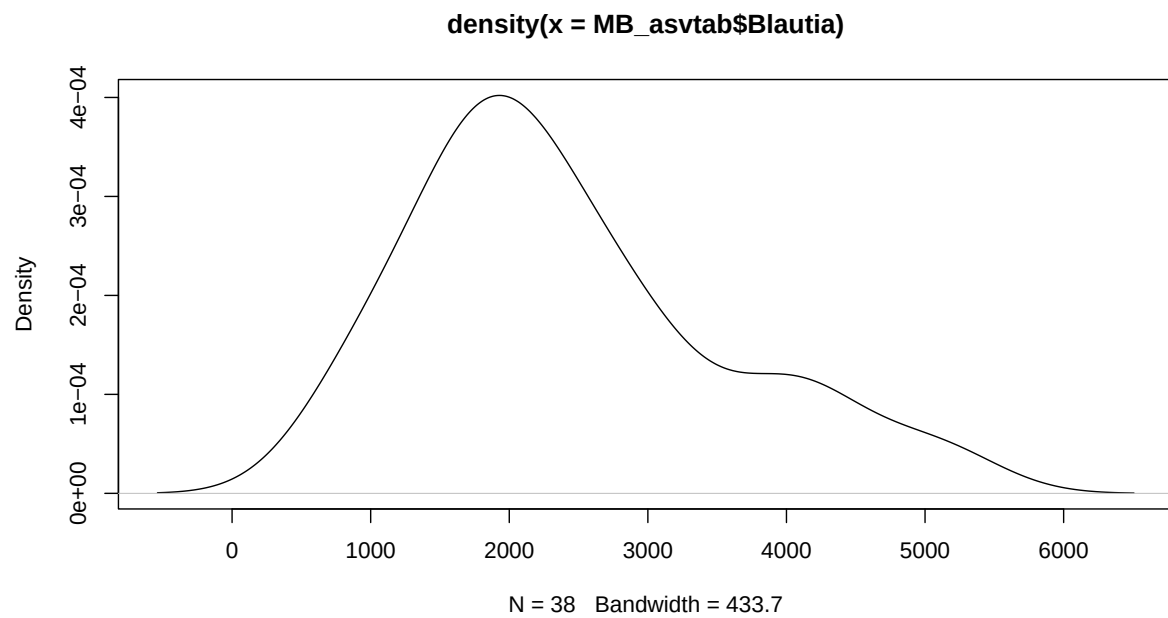
```
plot(density(MB_asvtab$Ruminococcus))
```



```
plot(density(MB_asvtab$Roseburia))
```



```
plot(density(MB_asvtab$Blautia))
```

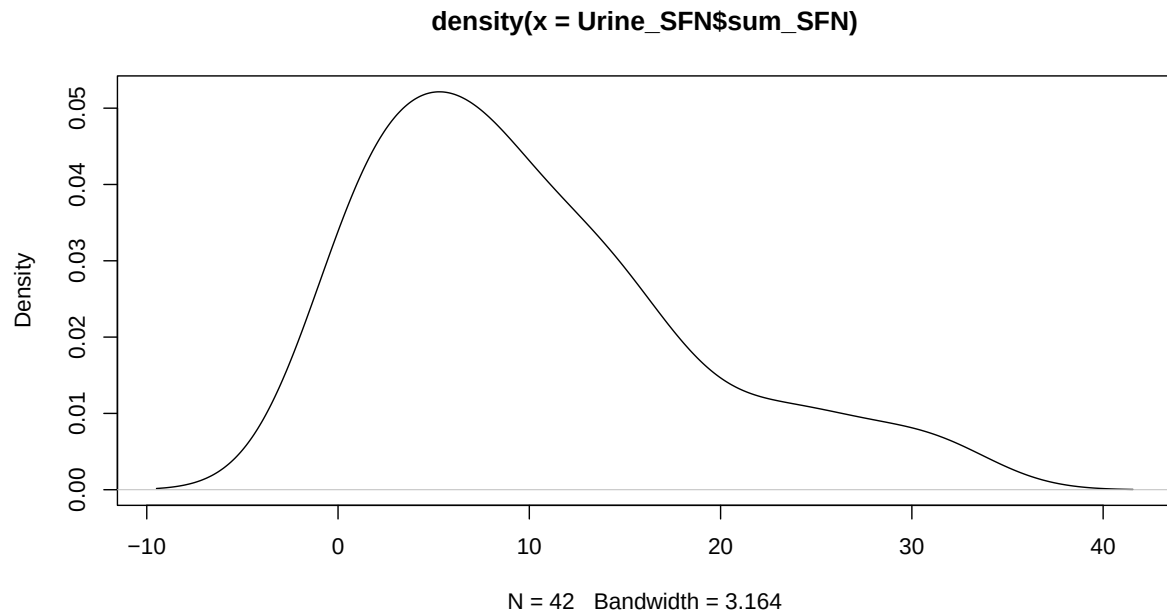


```
d <- density(MB_asvtab$Blautia)
quantile(MB_asvtab$Blautia)
##      0%      25%      50%      75%     100%
##  763.00 1680.50 2099.50 3017.25 5206.00
d$x[which.max(d$y)] # use the max peak of the curve for treat.value in mediate() later
## [1] 1929.751
```

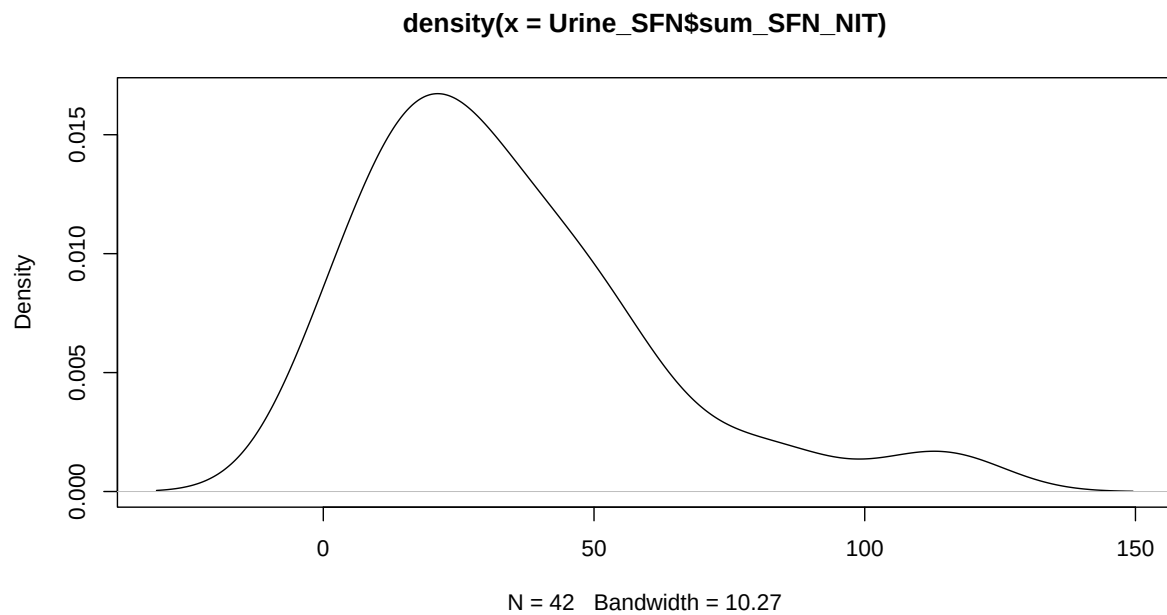
```
# this value is 1929.751
```

```
# most have right skewed density distributions with long right side tails, as is usual for ASV abundanc
```

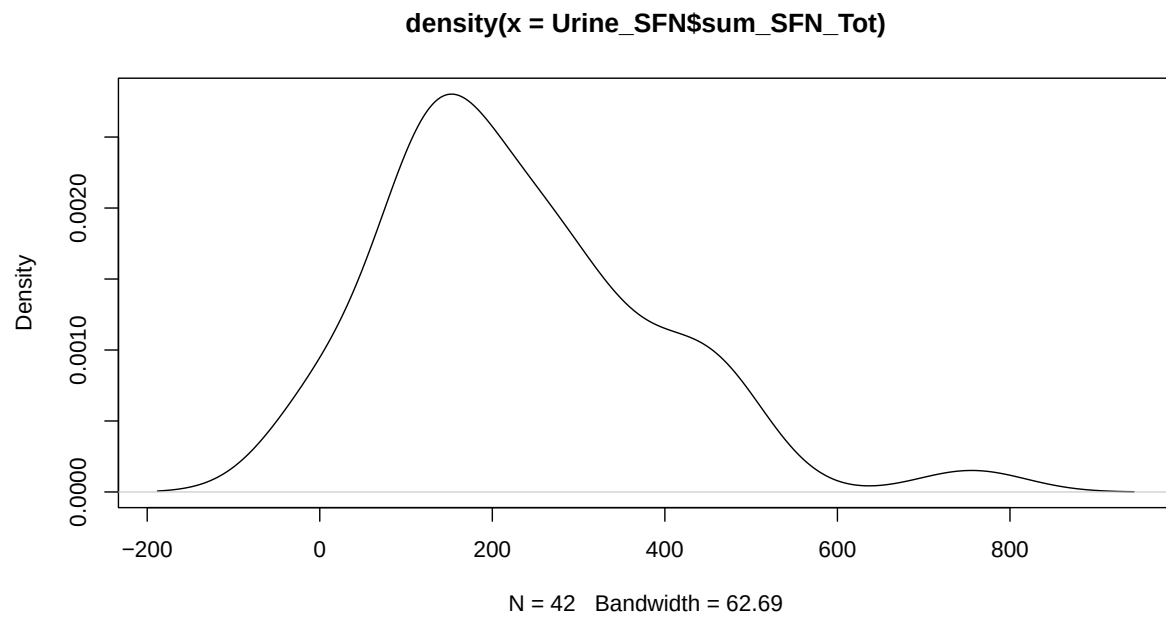
```
plot(density(Urine_SF_N$sum_SF_N)) # kinda right skewed
```



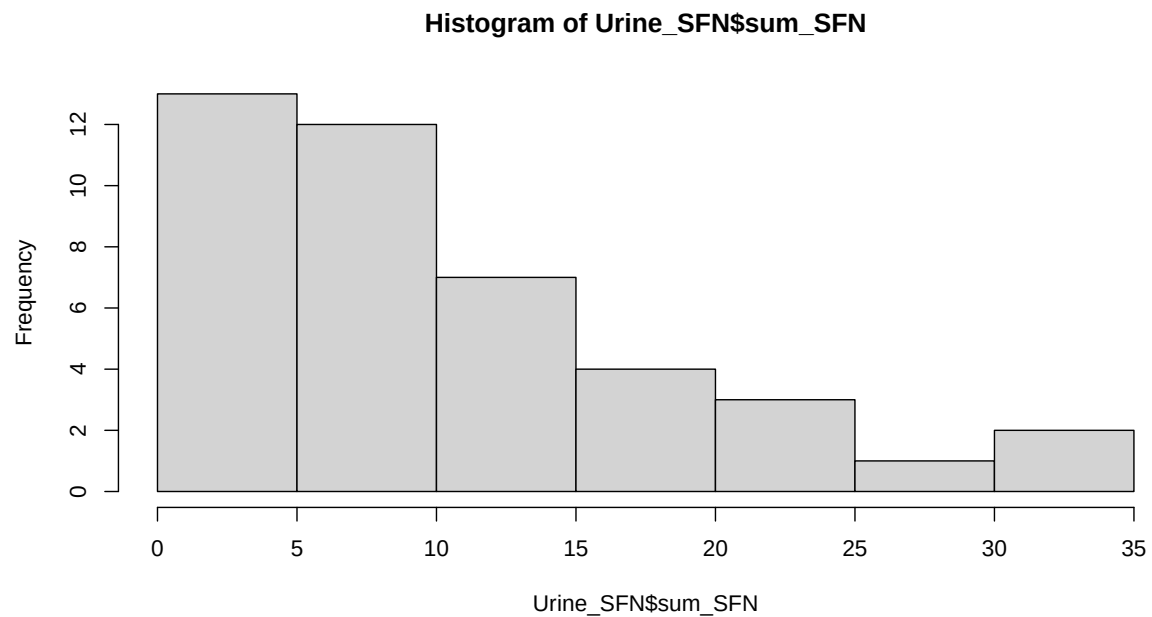
```
plot(density(Urine_SF_N$sum_SF_N_NIT)) # this one is more normal but has a right tail
```



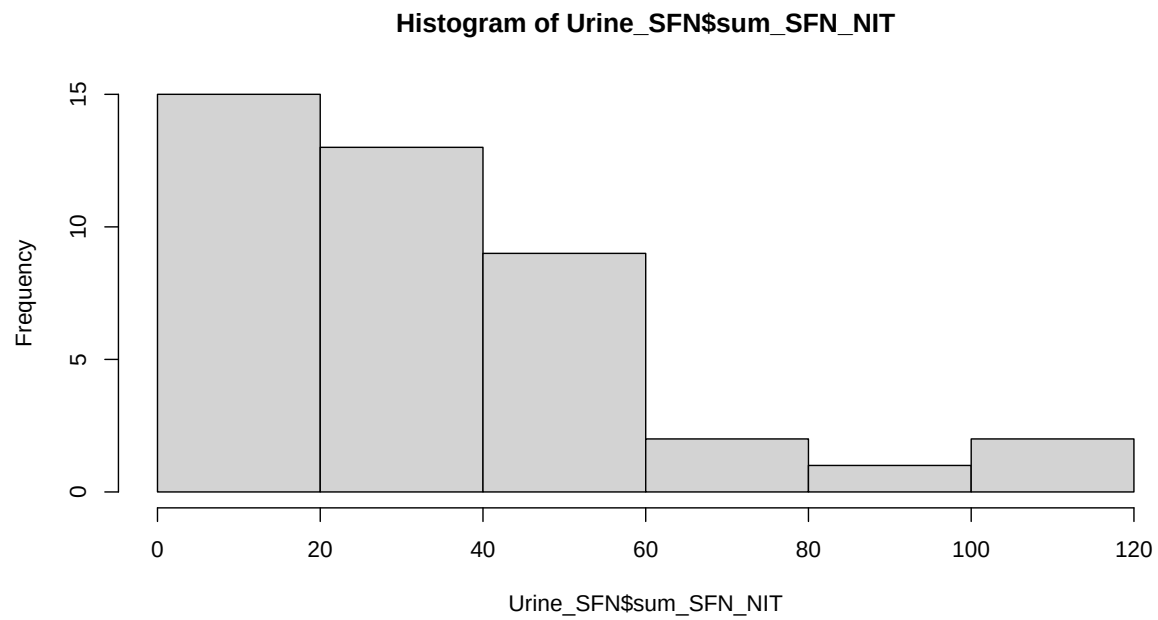
```
plot(density(Urine_SF_N$sum_SF_N_Tot)) # kinda right skewed
```



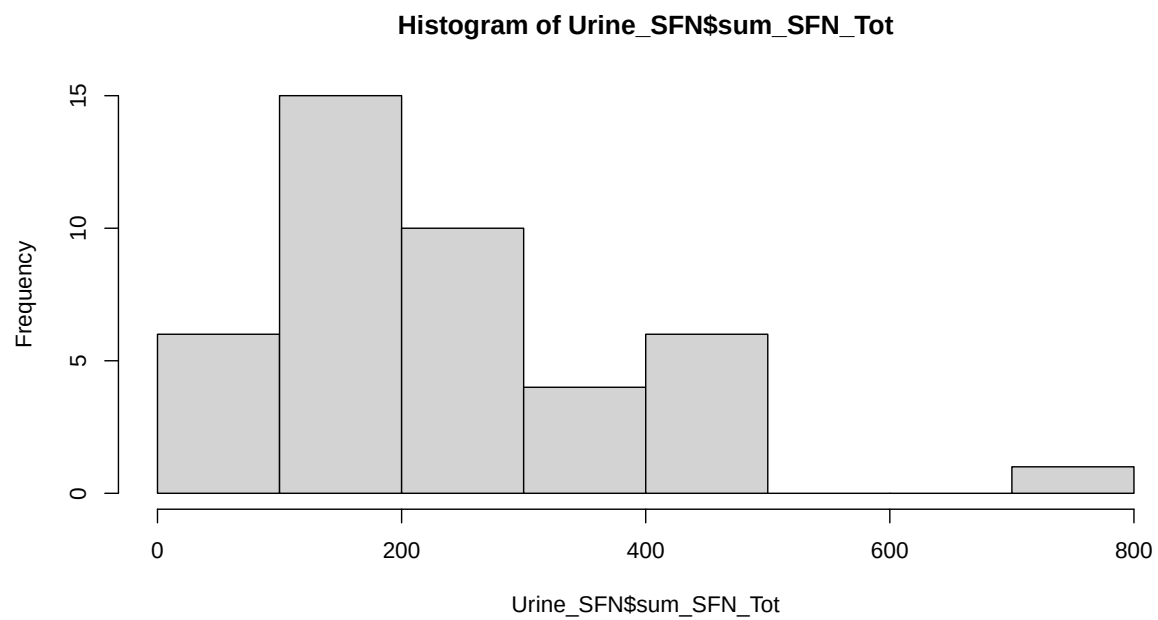
```
hist(Urine_SF_N$sum_SF_N)
```



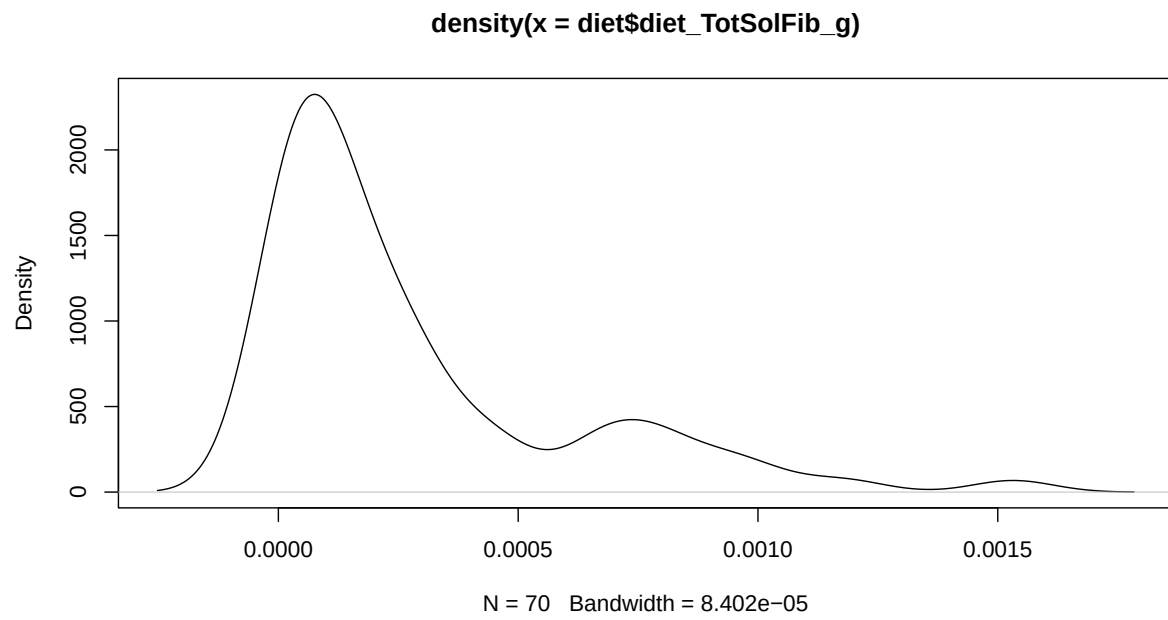
```
hist(Urine_SF_N$sum_SF_N_NIT)
```



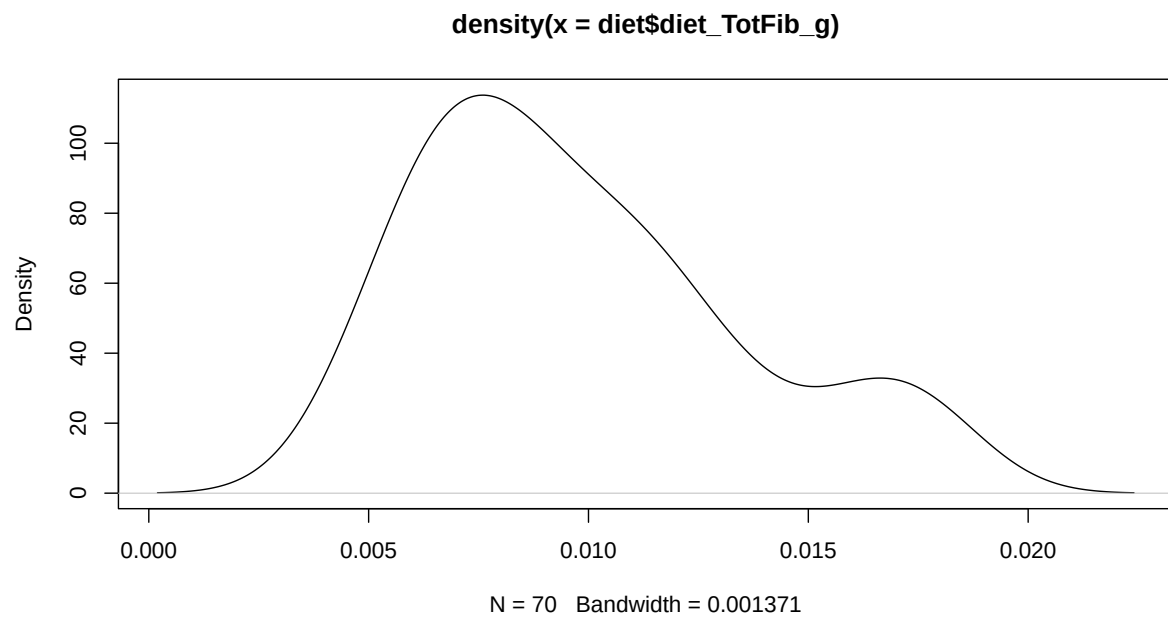
```
hist(Urine_SF_N$sum_SF_N_Tot)
```



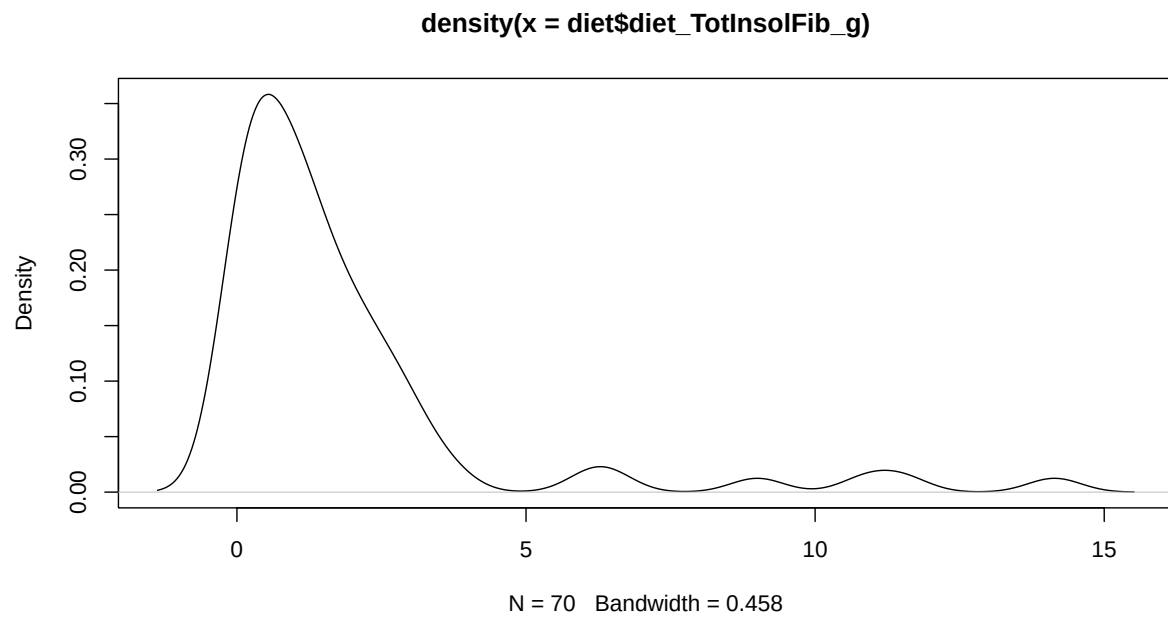
```
plot(density(diet$diet_TotSolFib_g)) # very right skewed
```



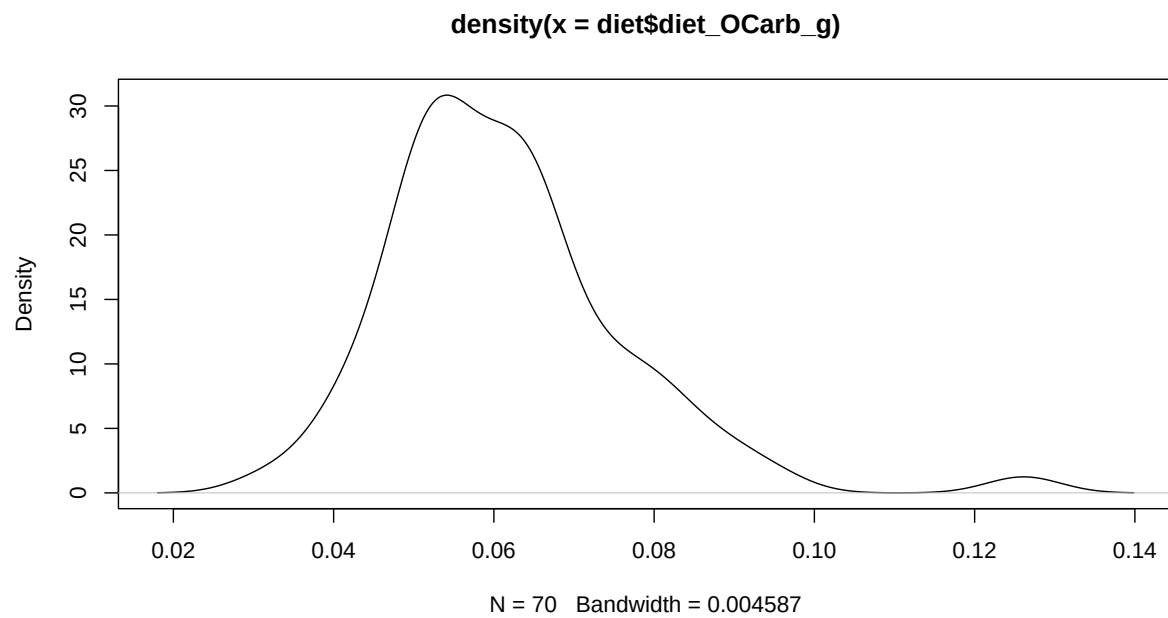
```
plot(density(diet$diet_TotFib_g)) # more normal
```



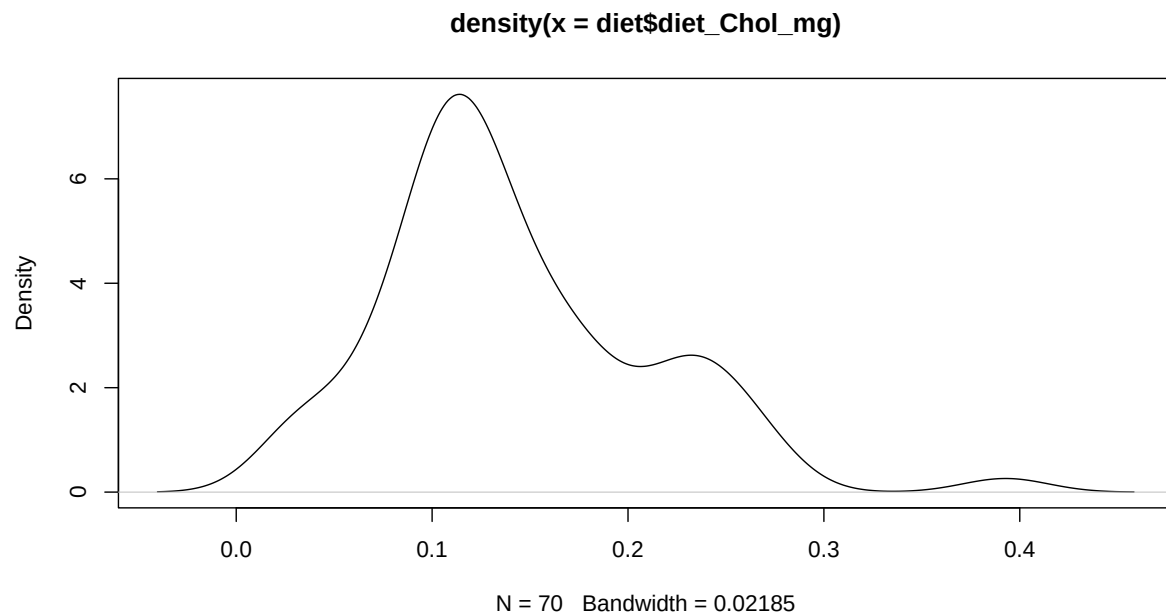
```
plot(density(diet$diet_TotInsolFib_g)) # right skewed
```



```
plot(density(diet$diet_OCarb_g)) # more normal but still slightly right skewed
```



```
plot(density(diet$diet_Chol_mg)) # more normal but still slightly right skewed
```



Run mediation analysis

- the hypothetical model we tested was whether certain ASVs (identified via partial correlation analysis; ASVs correlated to SFN and SFN-NIT) mediate the SFN response (SFN and SFN-NIT) to diet (based on several dietary components, particularly with interest in fiber)

try on one set of each X, Y, and M, then make into for loop based on visual model here: <https://library.virginia.edu/data/articles/introduction-to-mediation-analysis>

X <- dietary components

Y <- SFN (3 types)

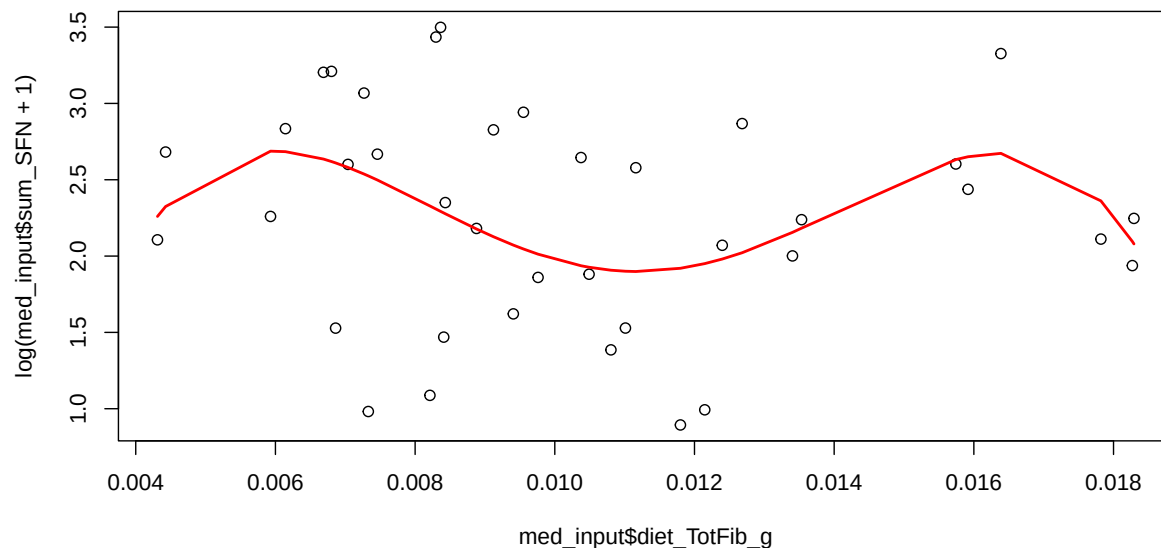
M <- ASVs

First, check model 0 is true

- will use this output to also limit how many mediations I run later (i.e., only model 0 combinations of SFN and diet components will move forward to mediation analysis)

```
# default lm
model.0.1 <- lm(sum_SFN ~ diet_TotFib_g, med_input)
# lm with log of SFN with pseudocount added
mod_log <- lm(log(sum_SFN+1) ~ diet_TotFib_g, med_input)
summary(mod_log)
##
## Call:
## lm(formula = log(sum_SFN + 1) ~ diet_TotFib_g, data = med_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -1.34576 -0.40833 0.03989 0.47716 1.19552
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.4579     0.3432   7.162 2.02e-08 ***
## diet_TotFib_g -18.5176    31.3780  -0.590   0.559
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7185 on 36 degrees of freedom
## Multiple R-squared:  0.009582, Adjusted R-squared:  -0.01793
## F-statistic: 0.3483 on 1 and 36 DF, p-value: 0.5588
plot(x = med_input$diet_TotFib_g, y = log(med_input$sum_SFN + 1))
# lm with polynomial
mod_logpoly <- lm(log(sum_SFN+1) ~ diet_TotFib_g + I(diet_TotFib_g^2) + I(diet_TotFib_g^3) + I(diet_TotFib_g^4), data = med_input)
summary(mod_logpoly)
##
## Call:
## lm(formula = log(sum_SFN + 1) ~ diet_TotFib_g + I(diet_TotFib_g^2) +
##      I(diet_TotFib_g^3) + I(diet_TotFib_g^4), data = med_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.54348 -0.34172  0.01183  0.55878  1.20569
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.518e+00  5.983e+00  -1.257   0.2177
## diet_TotFib_g     4.664e+03  2.553e+03   1.827   0.0767 .
## I(diet_TotFib_g^2) -7.415e+05  3.844e+05  -1.929   0.0624 .
## I(diet_TotFib_g^3)  4.767e+07  2.425e+07   1.966   0.0578 .
## I(diet_TotFib_g^4) -1.066e+09  5.432e+08  -1.963   0.0582 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6969 on 33 degrees of freedom
## Multiple R-squared:  0.1457, Adjusted R-squared:  0.04215
## F-statistic: 1.407 on 4 and 33 DF, p-value: 0.2532
plot(x = med_input$diet_TotFib_g, y = log(med_input$sum_SFN + 1))
pred <- predict(mod_logpoly)
ix <- sort(med_input$diet_TotFib_g, index.return = T)$ix
lines(med_input$diet_TotFib_g[ix], pred[ix], col='red', lwd = 2)
```



```
# gamma glm with pseudocount added to SFN
mod_gam <- glm((sum_SFN+1) ~ diet_TotFib_g, med_input, family = Gamma)
summary(mod_gam)
##
## Call:
## glm(formula = (sum_SFN + 1) ~ diet_TotFib_g, family = Gamma,
##      data = med_input)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.05965    0.02604   2.291  0.0279 *
## diet_TotFib_g 2.29669    2.55339   0.899  0.3744
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.4449801)
##
##      Null deviance: 17.346  on 37  degrees of freedom
## Residual deviance: 16.971  on 36  degrees of freedom
## AIC: 258.34
##
## Number of Fisher Scoring iterations: 5
# inverse gaussian flm with pseudocount added to SFN
mod_invgaus <- glm((sum_SFN+1) ~ diet_TotFib_g, med_input,
                  family = inverse.gaussian(link = "log"))
summary(mod_invgaus)
##
## Call:
## glm(formula = (sum_SFN + 1) ~ diet_TotFib_g, family = inverse.gaussian(link = "log"),
##      data = med_input)
##
```

```

## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.7325     0.3197   8.548 3.44e-10 ***
## diet_TotFib_g -23.4959    28.3264  -0.829   0.412
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for inverse.gaussian family taken to be 0.03690452)
##
##      Null deviance: 1.9252  on 37  degrees of freedom
## Residual deviance: 1.8977  on 36  degrees of freedom
## AIC: 258.44
##
## Number of Fisher Scoring iterations: 7
# tweedie
mod_tw <- glm((sum_SFN+1) ~ diet_TotFib_g, med_input,
              family = tweedie(var.power = 1, link.power = 0))
summary(mod_tw) # deviance super high
##
## Call:
## glm(formula = (sum_SFN + 1) ~ diet_TotFib_g, family = tweedie(var.power = 1,
##      link.power = 0), data = med_input)
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.7691     0.3183   8.700 2.23e-10 ***
## diet_TotFib_g -27.0621    30.1536  -0.897   0.375
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Tweedie family taken to be 5.396297)
##
##      Null deviance: 189.43  on 37  degrees of freedom
## Residual deviance: 184.98  on 36  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
# negative binomial
mod_negbin <- glm.nb((sum_SFN+1) ~ diet_TotFib_g, med_input)
summary(mod_negbin)
##
## Call:
## glm.nb(formula = (sum_SFN + 1) ~ diet_TotFib_g, data = med_input,
##      init.theta = 3.046952046, link = log)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.7537     0.3063   8.991 <2e-16 ***
## diet_TotFib_g -25.5397    28.2008  -0.906   0.365
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(3.047) family taken to be 1)

```

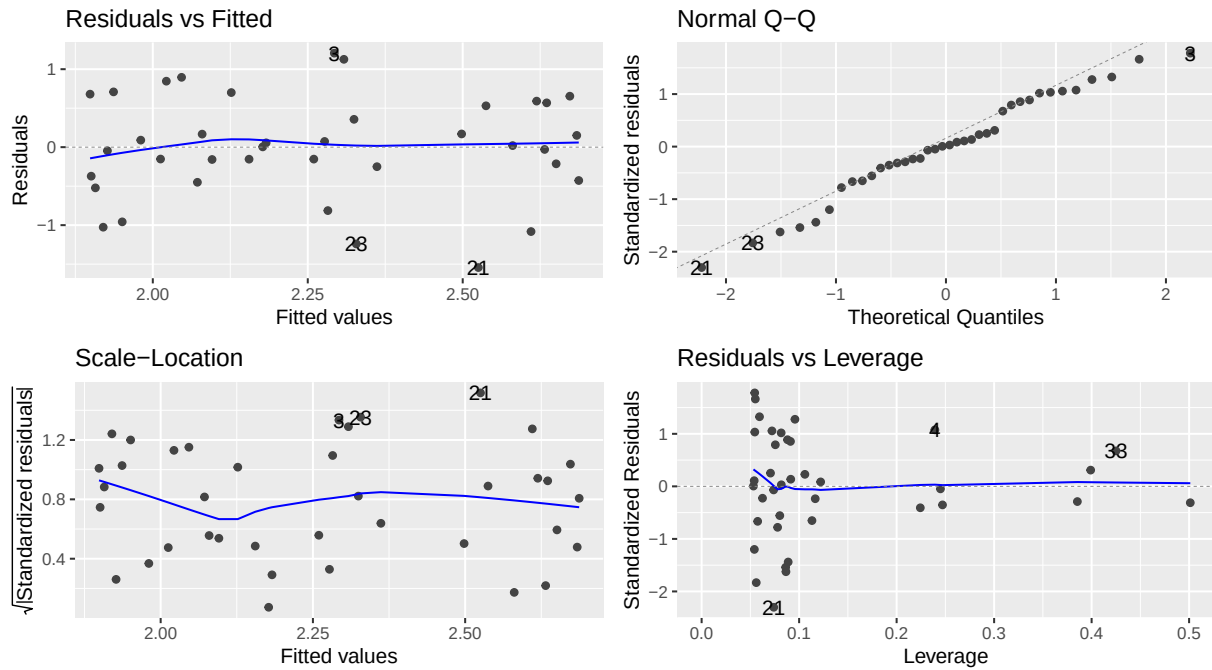
```
##
##      Null deviance: 40.028  on 37  degrees of freedom
## Residual deviance: 39.169  on 36  degrees of freedom
## AIC: 259.52
##
## Number of Fisher Scoring iterations: 1
##
##
##           Theta:  3.047
##          Std. Err.:  0.860
##
## 2 x log-likelihood:  -253.519

# compare with aic
AIC(model.0.1, mod_log, mod_logpoly, mod_gam, mod_invgaus, mod_negbin)
##           df          AIC
## model.0.1      3 271.03013
## mod_log        3  86.65626
## mod_logpoly    6  87.03790
## mod_gam        3 258.34107
## mod_invgaus    3 258.43624
## mod_negbin     3 259.51943
# lm with log(SFN+1) is the best model, but the polynomial is close behind
# however, adjusted R^2 is higher in the logpoly model, so I should probably use that moving forward

# model.0 <- lm(Y ~ X)
model.0.1 <- lm(log(sum_SFN+1) ~ diet_TotFib_g + I(diet_TotFib_g^2) + I(diet_TotFib_g^3) + I(diet_TotFib_g^4), data = med_input)
summary(model.0.1)
##
## Call:
## lm(formula = log(sum_SFN + 1) ~ diet_TotFib_g + I(diet_TotFib_g^2) +
##      I(diet_TotFib_g^3) + I(diet_TotFib_g^4), data = med_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.54348 -0.34172  0.01183  0.55878  1.20569
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -7.518e+00  5.983e+00  -1.257   0.2177
## diet_TotFib_g    4.664e+03  2.553e+03   1.827   0.0767 .
## I(diet_TotFib_g^2) -7.415e+05  3.844e+05  -1.929   0.0624 .
## I(diet_TotFib_g^3)  4.767e+07  2.425e+07   1.966   0.0578 .
## I(diet_TotFib_g^4) -1.066e+09  5.432e+08  -1.963   0.0582 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6969 on 33 degrees of freedom
## Multiple R-squared:  0.1457, Adjusted R-squared:  0.04215
## F-statistic: 1.407 on 4 and 33 DF,  p-value: 0.2532
# every term is almost significant

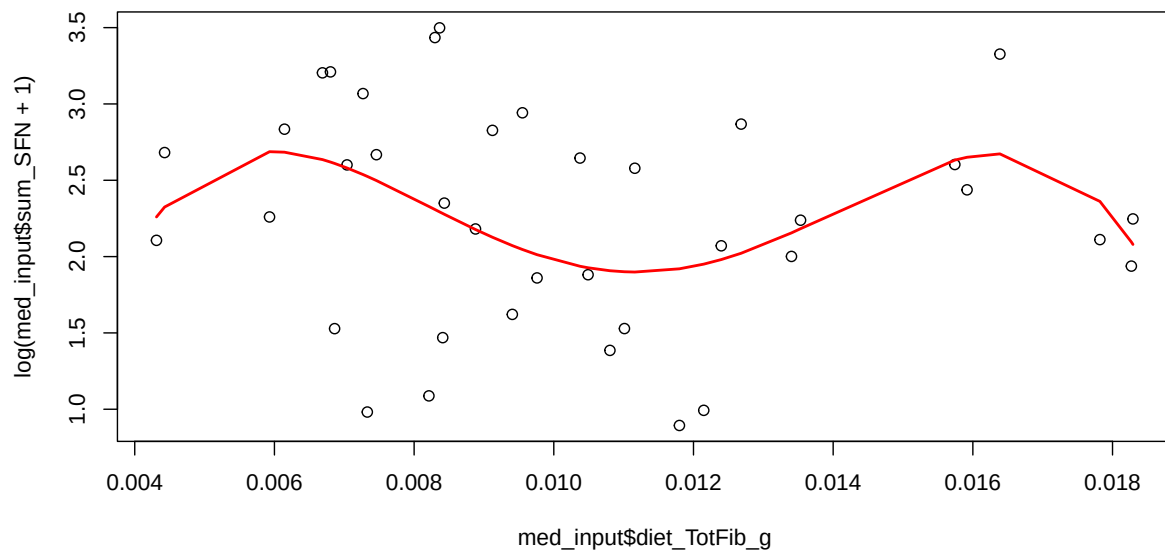
# diagnostic plots
```

```
autoplot(model.0.1)
```



Homoskedastic. Residuals are more or less normally distributed. Residuals are mostly linear.

```
# plot
plot(x = med_input$diet_TotFib_g, y = log(med_input$sum_SFN + 1))
pred <- predict(model.0.1)
ix <- sort(med_input$diet_TotFib_g, index.return = T)$ix
lines(med_input$diet_TotFib_g[ix], pred[ix], col='red', lwd = 2)
```



```

model.0.2 <- lm(log(sum_SFN_NIT+1) ~ diet_TotFib_g + I(diet_TotFib_g^2) + I(diet_TotFib_g^3) + I(diet_TotFib_g^4), data = med_input)
summary(model.0.2)
##
## Call:
## lm(formula = log(sum_SFN_NIT + 1) ~ diet_TotFib_g + I(diet_TotFib_g^2) +
##      I(diet_TotFib_g^3) + I(diet_TotFib_g^4), data = med_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.72027 -0.42765 -0.01992  0.45185  1.19609
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.170e+00  5.911e+00  -0.367   0.716
## diet_TotFib_g    2.530e+03  2.522e+03   1.003   0.323
## I(diet_TotFib_g^2) -3.770e+05  3.798e+05  -0.993   0.328
## I(diet_TotFib_g^3)  2.249e+07  2.396e+07   0.939   0.355
## I(diet_TotFib_g^4) -4.636e+08  5.367e+08  -0.864   0.394
##
## Residual standard error: 0.6886 on 33 degrees of freedom
## Multiple R-squared:  0.08722,    Adjusted R-squared:  -0.02342
## F-statistic: 0.7883 on 4 and 33 DF,  p-value: 0.5411
# no significance

```

```

list_sfn <- c("sum_SFN", "sum_SFN_Cys", "sum_SFN_NAC", "sum_SFN_CG",
              "sum_SFN_GSH", "sum_SFN_NIT", "sum_SFN_Tot")
# and use diet_compnts as the list of dietary components
# make an output file
model.0_lms <- c()
pvals_model.0_lms <- c()

for (a in 1:length(list_sfn)) {
  for (e in 1:length(diet_compnts)) {
    model.0 <- lm(paste0("log(", list_sfn[a], "+1) ~ ",
                        diet_compnts[e], "+ I(", diet_compnts[e], "^2) + I(",
                        diet_compnts[e], "^3) + I(", diet_compnts[e], "^4)",
                        data = med_input)

    modsum <- summary(model.0)
    coeff_tab <- as.data.frame(modsum$coefficients)
    model.0_lms[[paste(list_sfn[a], "~", diet_compnts[e])]] <- coeff_tab
    pvals_model.0_lms[[paste(list_sfn[a], "~", diet_compnts[e])]][["diet"]] <- coeff_tab[2,4]
    pvals_model.0_lms[[paste(list_sfn[a], "~", diet_compnts[e])]][["diet2"]] <- coeff_tab[3,4]
    pvals_model.0_lms[[paste(list_sfn[a], "~", diet_compnts[e])]][["diet3"]] <- coeff_tab[4,4]
    pvals_model.0_lms[[paste(list_sfn[a], "~", diet_compnts[e])]][["diet4"]] <- coeff_tab[5,4]
    e <- e + 1
  }
  a <- a + 1
}

pvals_model.0_lms <- t(as.data.frame(pvals_model.0_lms))
# a few are significant with an alpha of 0.05

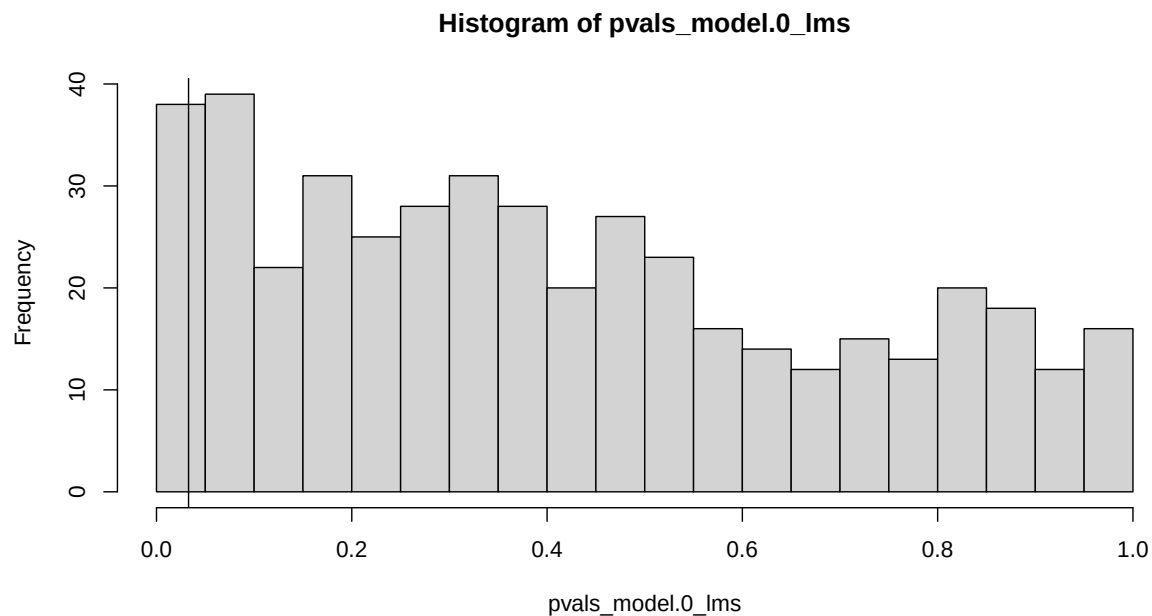
```

```

# want to save insoluble fiber to show coauthors and maybe reviewers
justinsols <- as.data.frame(pvals_model.0_lms) %>%
  filter(grepl("diet_TotInsolFib_g", rownames(pvals_model.0_lms)))
# write.table(justinsols, "../outputs/insol_pvals.txt", quote = F)

hist(pvals_model.0_lms, breaks = 20)
abline(v = quantile(pvals_model.0_lms, 0.05))

```



```

# which ones are those?
sign_dietSFNcombos <- pvals_model.0_lms[which(pvals_model.0_lms <= 0.05), ]
sign_dietSFNcombos
##          sum_SFN...diet_Alc_g.diet
##                      0.040577890
##          sum_SFN...diet_Alc_g.diet2
##                      0.041217022
##          sum_SFN...diet_Vit_B1_mg.diet
##                      0.011506227
##          sum_SFN...diet_Vit_B1_mg.diet2
##                      0.008644618
##          sum_SFN...diet_Vit_B1_mg.diet3
##                      0.006931764
##          sum_SFN...diet_Vit_B1_mg.diet4
##                      0.006240444
##          sum_SFN...diet_Iron_mg.diet
##                      0.013854151
##          sum_SFN...diet_Iron_mg.diet2
##                      0.015672449
##          sum_SFN...diet_Iron_mg.diet3
##                      0.018110412
##          sum_SFN...diet_Iron_mg.diet4
##                      0.021032034

```

```

##          sum_SFN_Cys...diet_Alc_g.diet
##          0.002241525
##          sum_SFN_Cys...diet_Alc_g.diet2
##          0.010586294
##          sum_SFN_Cys...diet_Alc_g.diet3
##          0.049564738
##          sum_SFN_Cys...diet_TotSolFib_g.diet
##          0.038840661
##          sum_SFN_Cys...diet_Fol.DFE_mcg_DFE.diet
##          0.023293845
##          sum_SFN_Cys...diet_Fol.DFE_mcg_DFE.diet2
##          0.021059696
##          sum_SFN_Cys...diet_Fol.DFE_mcg_DFE.diet3
##          0.021803070
##          sum_SFN_Cys...diet_Fol.DFE_mcg_DFE.diet4
##          0.023830455
##          sum_SFN_NAC...diet_Alc_g.diet
##          0.003842307
##          sum_SFN_NAC...diet_Alc_g.diet2
##          0.010741173
##          sum_SFN_NAC...diet_Alc_g.diet3
##          0.043078672
##          sum_SFN_NAC...diet_TotFib_g.diet
##          0.046047626
##          sum_SFN_NAC...diet_TotFib_g.diet2
##          0.046620371
##          sum_SFN_CG...diet_Cholesterol_mg.diet
##          0.038950624
##          sum_SFN_CG...diet_Cholesterol_mg.diet2
##          0.041030579
##          sum_SFN_CG...diet_Alc_g.diet
##          0.018003057
##          sum_SFN_CG...diet_Alc_g.diet2
##          0.049165400
##          sum_SFN_GSH...diet_Vit_B1_mg.diet
##          0.010773993
##          sum_SFN_GSH...diet_Vit_B1_mg.diet2
##          0.017851640
##          sum_SFN_GSH...diet_Vit_B1_mg.diet3
##          0.025393989
##          sum_SFN_GSH...diet_Vit_B1_mg.diet4
##          0.031546921
##          sum_SFN_Tot...diet_Alc_g.diet
##          0.003884531
##          sum_SFN_Tot...diet_Alc_g.diet2
##          0.010479773
##          sum_SFN_Tot...diet_Alc_g.diet3
##          0.044113371
##          sum_SFN_Tot...diet_Fol.DFE_mcg_DFE.diet
##          0.038037784
##          sum_SFN_Tot...diet_Fol.DFE_mcg_DFE.diet2
##          0.035250927
##          sum_SFN_Tot...diet_Fol.DFE_mcg_DFE.diet3

```

```
##                                0.037257742
## sum_SFN_Tot...diet_Fol.DFE_mcg_DFE.diet4
##                                0.041233040

# there are a ton of these that are significant
```

Pick model 0 as SFN and total dietary fiber to try code on

Troubleshoot model Y and M that will go into mediation

```
## try to use a tweedie model
model.Y.1 <- glm(sum_SFN ~ diet_TotFib_g + Blautia, med_input,
  family = tweedie(var.power = 1, link.power = 0))
# no warnings
summary(model.Y.1)
# its ok, no warnings now, but the deviance has gone up, no AIC, and more overdispersed

## try scaling sfn
# med_input$SFN_scaled <- round(((med_input$sum_SFN - min(med_input$sum_SFN)) / (max(med_input$sum_SFN))
# plot(density(med_input$SFN_scaled))
# plot(density(med_input$sum_SFN))
#
# model.Y.2 <- glm.nb(SFN_scaled ~ diet_TotFib_g + Blautia, med_input)
# summary(model.Y.2)
# plot(x = med_input$Blautia, y = med_input$SFN_scaled)
# # the same as unscaled so no point in scaling

## try log transforming
model.Y.3 <- glm.nb(log(sum_SFN+1) ~ diet_TotFib_g + Blautia, med_input)
summary(model.Y.3) # nothing
model.Y.3b <- glm.nb(sum_SFN ~ diet_TotFib_g + Blautia, med_input)

## try lm
model.Y.4 <- lm(log(sum_SFN+1) ~ diet_TotFib_g + log(Blautia+1), med_input)
summary(model.Y.4) # nothing

## inverse gaussian
model.Y.5 <- glm(formula = log(sum_SFN + 1) ~ diet_TotFib_g + Blautia,
  family = inverse.gaussian(link = "log"), data = med_input)
summary(model.Y.5)

## inv gaussian wo log sfn
model.Y.5b <- glm(formula = sum_SFN ~ diet_TotFib_g + Blautia,
  family = inverse.gaussian(link = "log"), data = med_input)

## use AIC to evaluate best option
AIC(model.Y.0, model.Y.2, model.Y.3, model.Y.3b, model.Y.4, model.Y.5, model.Y.5b)
AICtweedie( model.Y.1, dispersion = 1, k = 2, verbose = TRUE) # so really not a good fit at all for the
# compare

# model Y4 has a negative adj R2 despite having the lowest AIC, so Y5 is the best choice here
```

```

plot(med_input$quarts_totfib, med_input$f_Erysipelotrichaceae_ASV487)
plot(med_input$quarts_totfib, log(med_input$f_Erysipelotrichaceae_ASV487 + 1))

## basic lm
model.M.0 <- lm(f_Erysipelotrichaceae_ASV487 ~ diet_TotFib_g, med_input)
summary(model.M.0)

## glm negative binomial
model.M.1 <- glm.nb(f_Erysipelotrichaceae_ASV487 ~ diet_TotFib_g, med_input)
summary(model.M.1)

## log lm
model.M.2 <- lm(log(f_Erysipelotrichaceae_ASV487+1) ~ diet_TotFib_g, med_input)
summary(model.M.2)

## zero-infl neg bin
library(pscl)
model.M.3 <- zeroinfl(f_Erysipelotrichaceae_ASV487 ~ diet_TotFib_g, med_input,
  dist = "negbin")
summary(model.M.3)

## evaluate
AIC(model.M.0, model.M.1, model.M.2, model.M.3)
# model M2 is best (tried with a couple ASVs)

# add quartiles for X (diet_TotFib_g)
med_input <- med_input %>%
  mutate(quarts_totfib = ntile(diet_TotFib_g, 4),
    SFN_trans = log(sum_SFN+1),
    SFN_Tot_trans = log(sum_SFN_Tot+1),
    Blautia_trans = log(Blautia))
model.M <- lm(Blautia_trans ~ quarts_totfib, med_input)
model.Y <- lm(SFN_trans ~ quarts_totfib + Blautia, med_input)

results <- mediate(model.M, model.Y, treat = "quarts_totfib", mediator = "Blautia",
  control.value = 1, treat.value = 4, # lowest and highest quartiles
  boot = TRUE, sims = 500)
summary(results) # not great but at least now I have a way to implement loop code
##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##
##      Estimate 95% CI Lower 95% CI Upper p-value
## ACME      -1.81e-05   -1.17e-04      0.00   0.62
## ADE       -2.05e-01   -7.69e-01      0.35   0.48
## Total Effect -2.05e-01   -7.69e-01      0.35   0.48
## Prop. Mediated 8.85e-05   -1.70e-03      0.00   0.91
##
## Sample Size Used: 38
##
##
## Simulations: 500

```

Implement loop to cycle through each relevant SFN-dietary combo and each ASV

- only use significant SFN-diet combos from above

```
# Use list_sfn and diet_compnts as the list of dietary components
# make SFN taxa a list
taxa_list <- as.character(SFN_taxa$x)
# make an output file
med_res <- c()
pvals_med <- c()
exp_res_med <- c()
list_combos <- as.data.frame(sign_dietSFNcombos) %>%
  rownames_to_column("comparison") %>%
  dplyr::filter(!grepl("diet2|diet3|diet4", comparison)) %>%
  mutate(comparison = str_remove(comparison, ".diet$"),
         sfn = stringr::str_split(comparison, "\\...", simplify = T)[, 1],
         diet = stringr::str_split(comparison, "\\...", simplify = T)[, 2],
         diet = replace(diet, diet == "diet_Fol", "diet_Fol.DFE_mcg_DFE")) %>%
  dplyr::select(-c("comparison", "sign_dietSFNcombos"))

for (a in 1:nrow(list_combos)) {
  for (f in 1:length(taxa_list)) {
    # add quartiles for each dietary component that will be used
    # transform ASV abundance and transform SFN abundance
    med_input <- med_input %>%
      dplyr::mutate(quarts = ntile(med_input[[list_combos[a,2]]], 4),
                   sfn_trans = log(med_input[[list_combos[a,1]]] + 1),
                   asv_trans = log(med_input[[taxa_list[f]]] + 1),
                   asv = med_input[[taxa_list[f]]])

    # set models
    model.M <- lm(asv_trans ~ quarts,
                  med_input)
    model.Y <- lm(sfn_trans ~ quarts + asv_trans,
                  med_input)

    # mediate
    results <- mediate(model.M, model.Y,
                      treat = "quarts",
                      mediator = "asv_trans",
                      control.value = 1, treat.value = 4, # lowest and highest quartiles
                      boot = TRUE, sims = 500)

    # save results
    exp_res_med[[paste(list_combos[a,1], taxa_list[f], list_combos[a,2])] <- results
    modsum <- summary(results)
    pvals_med[[paste(list_combos[a,1], taxa_list[f], list_combos[a,2])] <- list(ACME_pval = modsum$d,
                                         ADE_pval = modsum$z.avg.p,
                                         prop_mediated_pval = modsum$prop.mediated.p)

    modsum$p.adj <- p.adjust(modsum$d.avg.p, method = "fdr")

    if (modsum$p.adj <= 0.2) {
      coeff_tab <- list(ACME_est = modsum$d.avg,
```

```

    ACME_pval = modsum$d.avg.p,
    ADE_est = modsum$z.avg,
    ADE_pval = modsum$z.avg.p,
    prop_mediated_est = modsum$n.avg,
    prop_mediated_pval = modsum$n.avg.p)

    med_res[[paste(list_combos[a,1], taxa_list[f], list_combos[a,2])] <- coeff_tab

  } else { print(paste(list_combos[a,1], ",", taxa_list[f], ",",
                        list_combos[a,2], "are NOT significant"))}

  # counters
  f <- f + 1
}
a <- a + 1
}

## [1] "sum_SFN , Agathobacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Blautia , diet_Alc_g are NOT significant"
## [1] "sum_SFN , CAG.56 , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Colidextribacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Flavonifractor , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Incertae.Sedis , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Lachnospiraceae.NK4A136.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Turicibacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN , UCG.003 , diet_Alc_g are NOT significant"
## [1] "sum_SFN , X.Eubacterium..fissicatena.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN , X.Eubacterium..xylanophilum.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN , X.Ruminococcus..gnavreuii.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN , f_Erysipelotrichaceae_ASV487 , diet_Alc_g are NOT significant"
## [1] "sum_SFN , f_NA_ASV516 , diet_Alc_g are NOT significant"
## [1] "sum_SFN , f_UCG.010_ASV667 , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Holdemania , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Ruminococcus , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Subdoligranulum , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Alistipes , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Collinsella , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Faecalibacterium , diet_Alc_g are NOT significant"
## [1] "sum_SFN , NK4A214.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Roseburia , diet_Alc_g are NOT significant"
## [1] "sum_SFN , Agathobacter , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Blautia , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , CAG.56 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Colidextribacter , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Flavonifractor , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Incertae.Sedis , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Lachnospiraceae.NK4A136.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Turicibacter , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , UCG.003 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , X.Eubacterium..fissicatena.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , X.Eubacterium..xylanophilum.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , X.Ruminococcus..gnavreuii.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , X.Ruminococcus..gnavus.group , diet_Vit_B1_mg are NOT significant"

```

```

## [1] "sum_SFN , f_Erysipelotrichaceae_ASV487 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , f_NA_ASV516 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , f_UCG.010_ASV667 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Holdemania , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Ruminococcus , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Subdoligranulum , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Alistipes , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Bifidobacterium , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Faecalibacterium , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , NK4A214.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Roseburia , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN , Agathobacter , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Blautia , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , CAG.56 , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Colidextribacter , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Flavonifractor , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Incertae.Sedis , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Lachnospiraceae.NK4A136.group , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Turicibacter , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , UCG.003 , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , X.Eubacterium..fissicatena.group , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , X.Eubacterium..xylanophilum.group , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , X.Ruminococcus..gnavus.group , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , X.Ruminococcus..gnavus.group , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , f_Erysipelotrichaceae_ASV487 , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , f_NA_ASV516 , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , f_UCG.010_ASV667 , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Holdemania , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Ruminococcus , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Subdoligranulum , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Alistipes , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Bifidobacterium , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Collinsella , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Faecalibacterium , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , NK4A214.group , diet_Iron_mg are NOT significant"
## [1] "sum_SFN , Roseburia , diet_Iron_mg are NOT significant"
## [1] "sum_SFN_Cys , Agathobacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Blautia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , CAG.56 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Colidextribacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Flavonifractor , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Incertae.Sedis , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Lachnospiraceae.NK4A136.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Turicibacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , UCG.003 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , X.Eubacterium..fissicatena.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , X.Eubacterium..xylanophilum.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , f_Erysipelotrichaceae_ASV487 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , f_NA_ASV516 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , f_UCG.010_ASV667 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Holdemania , diet_Alc_g are NOT significant"

```

```

## [1] "sum_SFN_Cys , Ruminococcus , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Subdoligranulum , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Alistipes , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Bifidobacterium , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Collinsella , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , NK4A214.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Roseburia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Cys , Agathobacter , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Blautia , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , CAG.56 , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Colidextribacter , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Flavonifractor , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Incertae.Sedis , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Turicibacter , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , UCG.003 , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , X.Eubacterium..fissicatena.group , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , X.Eubacterium..xylanophilum.group , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , X.Ruminococcus..gnavreuii.group , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , X.Ruminococcus..gnavus.group , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , f_Erysipelotrichaceae_ASV487 , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , f_NA_ASV516 , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , f_UCG.010_ASV667 , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Holdemania , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Ruminococcus , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Subdoligranulum , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Alistipes , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Bifidobacterium , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Collinsella , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , NK4A214.group , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Roseburia , diet_TotSolFib_g are NOT significant"
## [1] "sum_SFN_Cys , Agathobacter , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Blautia , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , CAG.56 , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Colidextribacter , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Flavonifractor , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Incertae.Sedis , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Lachnospiraceae.NK4A136.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Turicibacter , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , UCG.003 , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , X.Eubacterium..fissicatena.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , X.Eubacterium..xylanophilum.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , X.Ruminococcus..gnavus.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , f_Erysipelotrichaceae_ASV487 , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , f_UCG.010_ASV667 , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Holdemania , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Ruminococcus , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Subdoligranulum , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Alistipes , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Bifidobacterium , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Collinsella , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Faecalibacterium , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , NK4A214.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Cys , Roseburia , diet_Fol.DFE_mcg_DFE are NOT significant"

```

```

## [1] "sum_SFN_NAC , Agathobacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Blautia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , CAG.56 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Colidextribacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Flavonifractor , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Incertae.Sedis , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Lachnospiraceae.NK4A136.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Turicibacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , UCG.003 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , X.Eubacterium..fissicatena.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , X.Eubacterium..xylanophilum.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , X.Ruminococcus..gavreauii.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , f_Erysipelotrichaceae_ASV487 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , f_NA_ASV516 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , f_UCG.010_ASV667 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Holdemania , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Ruminococcus , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Subdoligranulum , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Alistipes , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Bifidobacterium , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Collinsella , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , NK4A214.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Roseburia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_NAC , Agathobacter , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Blautia , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , CAG.56 , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Colidextribacter , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Flavonifractor , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Incertae.Sedis , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Lachnospiraceae.NK4A136.group , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Turicibacter , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , UCG.003 , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , X.Eubacterium..fissicatena.group , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , X.Eubacterium..xylanophilum.group , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , X.Ruminococcus..gavreauii.group , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , X.Ruminococcus..gnavus.group , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , f_Erysipelotrichaceae_ASV487 , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , f_NA_ASV516 , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , f_UCG.010_ASV667 , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Holdemania , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Ruminococcus , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Subdoligranulum , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Alistipes , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Bifidobacterium , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Faecalibacterium , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , NK4A214.group , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_NAC , Roseburia , diet_TotFib_g are NOT significant"
## [1] "sum_SFN_CG , Agathobacter , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Blautia , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , CAG.56 , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Colidextribacter , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Flavonifractor , diet_Chol_mg are NOT significant"

```

```

## [1] "sum_SFN_CG , Incertae.Sedis , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Lachnospiraceae.NK4A136.group , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Turicibacter , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , UCG.003 , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , X.Eubacterium..fissicatena.group , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , X.Eubacterium..xylanophilum.group , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , X.Ruminococcus..gnavus.group , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , X.Ruminococcus..gnavus.group , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , f_Erysipelotrichaceae_ASV487 , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , f_NA_ASV516 , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , f_UCG.010_ASV667 , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Holdemania , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Ruminococcus , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Alistipes , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Bifidobacterium , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Collinsella , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Faecalibacterium , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , NK4A214.group , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Roseburia , diet_Chol_mg are NOT significant"
## [1] "sum_SFN_CG , Agathobacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Blautia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , CAG.56 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Colidextribacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Flavonifractor , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Incertae.Sedis , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Lachnospiraceae.NK4A136.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Turicibacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , UCG.003 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , X.Eubacterium..fissicatena.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , X.Eubacterium..xylanophilum.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , f_Erysipelotrichaceae_ASV487 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , f_NA_ASV516 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , f_UCG.010_ASV667 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Holdemania , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Ruminococcus , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Subdoligranulum , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Alistipes , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Bifidobacterium , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Collinsella , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Faecalibacterium , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , NK4A214.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_CG , Roseburia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_GSH , Agathobacter , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Blautia , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , CAG.56 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Colidextribacter , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Flavonifractor , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Incertae.Sedis , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Lachnospiraceae.NK4A136.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Turicibacter , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , UCG.003 , diet_Vit_B1_mg are NOT significant"

```

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## [1] "sum_SFN_GSH , X.Eubacterium..fissicatena.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , X.Eubacterium..xylanophilum.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , X.Ruminococcus..gnavus.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , f_Erysipelotrichaceae_ASV487 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , f_NA_ASV516 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , f_UCG.010_ASV667 , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Holdemania , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Ruminococcus , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Subdoligranulum , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Alistipes , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Bifidobacterium , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Collinsella , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Faecalibacterium , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , NK4A214.group , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_GSH , Roseburia , diet_Vit_B1_mg are NOT significant"
## [1] "sum_SFN_Tot , Agathobacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Blautia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , CAG.56 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Colidextribacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Flavonifractor , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Incertae.Sedis , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Lachnospiraceae.NK4A136.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Turicibacter , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , UCG.003 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , X.Eubacterium..fissicatena.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , X.Eubacterium..xylanophilum.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , X.Ruminococcus..gnavus.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , f_Erysipelotrichaceae_ASV487 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , f_NA_ASV516 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , f_UCG.010_ASV667 , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Holdemania , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Ruminococcus , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Subdoligranulum , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Alistipes , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Bifidobacterium , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Collinsella , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , NK4A214.group , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Roseburia , diet_Alc_g are NOT significant"
## [1] "sum_SFN_Tot , Agathobacter , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Blautia , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , CAG.56 , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Colidextribacter , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Flavonifractor , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Incertae.Sedis , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Lachnospiraceae.NK4A136.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Turicibacter , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , UCG.003 , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , X.Eubacterium..fissicatena.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , X.Eubacterium..xylanophilum.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , X.Ruminococcus..gnavus.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , X.Ruminococcus..gnavus.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , f_Erysipelotrichaceae_ASV487 , diet_Fol.DFE_mcg_DFE are NOT significant"

```

```
## [1] "sum_SFN_Tot , Holdemania , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Ruminococcus , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Subdoligranulum , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Alistipes , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Bifidobacterium , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Faecalibacterium , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , NK4A214.group , diet_Fol.DFE_mcg_DFE are NOT significant"
## [1] "sum_SFN_Tot , Roseburia , diet_Fol.DFE_mcg_DFE are NOT significant"
```

```
# saveRDS(med_res, "mediation_result_notconservative.Rdata")
# saveRDS(pvals_med, "pvals_mediation_notconservative.Rdata")
# saveRDS(exp_res_med, "expanded_mediation_results_notconservative.Rdata")

# saveRDS(med_res, "mediation_result_signifsfndiets.Rdata")
# saveRDS(pvals_med, "pvals_mediation_signifsfndiets.Rdata")
# saveRDS(exp_res_med, "expanded_mediation_results_signifsfndiets.Rdata")

saveRDS(med_res, "mediation_result_signifcumSFNs.Rdata")
saveRDS(pvals_med, "pvals_mediation_signifcumSFNs.Rdata")
saveRDS(exp_res_med, "expanded_mediation_results_signifcumSFNs.Rdata")
```

Examine results

```
signif <- do.call(rbind.data.frame, pvals_med[names(med_res)]) %>%
  rownames_to_column("names(med_res)")
signif
##                               names(med_res) ACME_pval
## 1                sum_SFN Bifidobacterium diet_Alc_g    0.192
## 2                sum_SFN Collinsella diet_Vit_B1_mg    0.092
## 3                sum_SFN_Cys Faecalibacterium diet_Alc_g    0.196
## 4    sum_SFN_Cys Lachnospiraceae.NK4A136.group diet_TotSolFib_g    0.200
## 5                sum_SFN_Cys Faecalibacterium diet_TotSolFib_g    0.056
## 6    sum_SFN_Cys X.Ruminococcus..gavvreauii.group diet_Fol.DFE_mcg_DFE    0.144
## 7                sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE    0.128
## 8                sum_SFN_NAC Faecalibacterium diet_Alc_g    0.172
## 9                sum_SFN_NAC Collinsella diet_TotFib_g    0.072
## 10               sum_SFN_CG Subdoligranulum diet_Chol_mg    0.056
## 11    sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg    0.136
## 12               sum_SFN_Tot Faecalibacterium diet_Alc_g    0.164
## 13               sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE    0.116
## 14               sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE    0.152
## 15               sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE    0.112
##  ADE_pval prop_mediated_pval
## 1      0.116          0.528
## 2      0.240          0.684
## 3      0.540          0.436
## 4      0.284          0.604
## 5      0.908          0.544
## 6      0.772          0.828
## 7      0.672          0.900
## 8      0.960          0.692
```

```

## 9      0.828      0.684
## 10     0.032      0.216
## 11     0.676      0.384
## 12     0.804      0.852
## 13     0.696      0.796
## 14     0.980      0.736
## 15     0.336      0.828

# write.table(signif, "../outputs/signif_meds_cumsfn.txt")

med_res
## `$sum_SFN Bifidobacterium diet_Alc_g`
## `$sum_SFN Bifidobacterium diet_Alc_g`$ACME_est
## [1] 0.1367869
##
## `$sum_SFN Bifidobacterium diet_Alc_g`$ACME_pval
## [1] 0.192
##
## `$sum_SFN Bifidobacterium diet_Alc_g`$ADE_est
## [1] -0.3789608
##
## `$sum_SFN Bifidobacterium diet_Alc_g`$ADE_pval
## [1] 0.116
##
## `$sum_SFN Bifidobacterium diet_Alc_g`$prop_mediated_est
## [1] -0.5648293
##
## `$sum_SFN Bifidobacterium diet_Alc_g`$prop_mediated_pval
## [1] 0.528
##
##
## `$sum_SFN Collinsella diet_Vit_B1_mg`
## `$sum_SFN Collinsella diet_Vit_B1_mg`$ACME_est
## [1] -0.2192413
##
## `$sum_SFN Collinsella diet_Vit_B1_mg`$ACME_pval
## [1] 0.092
##
## `$sum_SFN Collinsella diet_Vit_B1_mg`$ADE_est
## [1] 0.3734043
##
## `$sum_SFN Collinsella diet_Vit_B1_mg`$ADE_pval
## [1] 0.24
##
## `$sum_SFN Collinsella diet_Vit_B1_mg`$prop_mediated_est
## [1] -1.422139
##
## `$sum_SFN Collinsella diet_Vit_B1_mg`$prop_mediated_pval
## [1] 0.684
##
##
## `$sum_SFN_Cys Faecalibacterium diet_Alc_g`
## `$sum_SFN_Cys Faecalibacterium diet_Alc_g`$ACME_est

```

```

## [1] 0.1239119
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_Alc\_g`$ACME\_pval
## [1] 0.196
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_Alc\_g`$ADE\_est
## [1] 0.2144161
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_Alc\_g`$ADE\_pval
## [1] 0.54
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_Alc\_g`$prop\_mediated\_est
## [1] 0.3662479
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_Alc\_g`$prop\_mediated\_pval
## [1] 0.436
##
##
##
## $\sum\_SFN\_Cys Lachnospiraceae.NK4A136.group diet\_TotSolFib\_g`
## $\sum\_SFN\_Cys Lachnospiraceae.NK4A136.group diet\_TotSolFib\_g`$ACME\_est
## [1] -0.1265578
##
## $\sum\_SFN\_Cys Lachnospiraceae.NK4A136.group diet\_TotSolFib\_g`$ACME\_pval
## [1] 0.2
##
## $\sum\_SFN\_Cys Lachnospiraceae.NK4A136.group diet\_TotSolFib\_g`$ADE\_est
## [1] 0.3242522
##
## $\sum\_SFN\_Cys Lachnospiraceae.NK4A136.group diet\_TotSolFib\_g`$ADE\_pval
## [1] 0.284
##
## $\sum\_SFN\_Cys Lachnospiraceae.NK4A136.group diet\_TotSolFib\_g`$prop\_mediated\_est
## [1] -0.6401692
##
## $\sum\_SFN\_Cys Lachnospiraceae.NK4A136.group diet\_TotSolFib\_g`$prop\_mediated\_pval
## [1] 0.604
##
##
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_TotSolFib\_g`
## $\sum\_SFN\_Cys Faecalibacterium diet\_TotSolFib\_g`$ACME\_est
## [1] 0.2260267
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_TotSolFib\_g`$ACME\_pval
## [1] 0.056
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_TotSolFib\_g`$ADE\_est
## [1] -0.02833233
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_TotSolFib\_g`$ADE\_pval
## [1] 0.908
##
## $\sum\_SFN\_Cys Faecalibacterium diet\_TotSolFib\_g`$prop\_mediated\_est
## [1] 1.143314

```

```

##
## `$sum_SFN_Cys Faecalibacterium diet_TotSolFib_g`$prop_mediated_pval
## [1] 0.544
##
##
##
## `$sum_SFN_Cys X.Ruminococcus..gavreaii.group diet_Fol.DFE_mcg_DFE`
## `$sum_SFN_Cys X.Ruminococcus..gavreaii.group diet_Fol.DFE_mcg_DFE`$ACME_est
## [1] 0.1523865
##
## `$sum_SFN_Cys X.Ruminococcus..gavreaii.group diet_Fol.DFE_mcg_DFE`$ACME_pval
## [1] 0.144
##
## `$sum_SFN_Cys X.Ruminococcus..gavreaii.group diet_Fol.DFE_mcg_DFE`$ADE_est
## [1] -0.09947036
##
## `$sum_SFN_Cys X.Ruminococcus..gavreaii.group diet_Fol.DFE_mcg_DFE`$ADE_pval
## [1] 0.772
##
## `$sum_SFN_Cys X.Ruminococcus..gavreaii.group diet_Fol.DFE_mcg_DFE`$prop_mediated_est
## [1] 2.879773
##
## `$sum_SFN_Cys X.Ruminococcus..gavreaii.group diet_Fol.DFE_mcg_DFE`$prop_mediated_pval
## [1] 0.828
##
##
##
## `$sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE`
## `$sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ACME_est
## [1] 0.1850021
##
## `$sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ACME_pval
## [1] 0.128
##
## `$sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ADE_est
## [1] -0.1320859
##
## `$sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ADE_pval
## [1] 0.672
##
## `$sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$prop_mediated_est
## [1] 3.496136
##
## `$sum_SFN_Cys f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$prop_mediated_pval
## [1] 0.9
##
##
##
## `$sum_SFN_NAC Faecalibacterium diet_Alc_g`
## `$sum_SFN_NAC Faecalibacterium diet_Alc_g`$ACME_est
## [1] 0.111018
##
## `$sum_SFN_NAC Faecalibacterium diet_Alc_g`$ACME_pval
## [1] 0.172
##
##
## `$sum_SFN_NAC Faecalibacterium diet_Alc_g`$ADE_est

```

```

## [1] -0.03740561
##
## $`sum_SFN_NAC Faecalibacterium diet_Alc_g`$ADE_pval
## [1] 0.96
##
## $`sum_SFN_NAC Faecalibacterium diet_Alc_g`$prop_mediated_est
## [1] 1.508143
##
## $`sum_SFN_NAC Faecalibacterium diet_Alc_g`$prop_mediated_pval
## [1] 0.692
##
##
## $`sum_SFN_NAC Collinsella diet_TotFib_g`
## $`sum_SFN_NAC Collinsella diet_TotFib_g`$ACME_est
## [1] -0.1419928
##
## $`sum_SFN_NAC Collinsella diet_TotFib_g`$ACME_pval
## [1] 0.072
##
## $`sum_SFN_NAC Collinsella diet_TotFib_g`$ADE_est
## [1] 0.05842425
##
## $`sum_SFN_NAC Collinsella diet_TotFib_g`$ADE_pval
## [1] 0.828
##
## $`sum_SFN_NAC Collinsella diet_TotFib_g`$prop_mediated_est
## [1] 1.699118
##
## $`sum_SFN_NAC Collinsella diet_TotFib_g`$prop_mediated_pval
## [1] 0.684
##
##
## $`sum_SFN_CG Subdoligranulum diet_Chol_mg`
## $`sum_SFN_CG Subdoligranulum diet_Chol_mg`$ACME_est
## [1] 0.02209599
##
## $`sum_SFN_CG Subdoligranulum diet_Chol_mg`$ACME_pval
## [1] 0.056
##
## $`sum_SFN_CG Subdoligranulum diet_Chol_mg`$ADE_est
## [1] -0.1135696
##
## $`sum_SFN_CG Subdoligranulum diet_Chol_mg`$ADE_pval
## [1] 0.032
##
## $`sum_SFN_CG Subdoligranulum diet_Chol_mg`$prop_mediated_est
## [1] -0.2415559
##
## $`sum_SFN_CG Subdoligranulum diet_Chol_mg`$prop_mediated_pval
## [1] 0.216
##
##
## $`sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg`

```

```

## `$sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg`$ACME_est
## [1] -0.00955483
##
## `$sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg`$ACME_pval
## [1] 0.136
##
## `$sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg`$ADE_est
## [1] -0.01034253
##
## `$sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg`$ADE_pval
## [1] 0.676
##
## `$sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg`$prop_mediated_est
## [1] 0.480206
##
## `$sum_SFN_GSH X.Ruminococcus..gnavus.group diet_Vit_B1_mg`$prop_mediated_pval
## [1] 0.384
##
##
## `$sum_SFN_Tot Faecalibacterium diet_Alc_g`
## `$sum_SFN_Tot Faecalibacterium diet_Alc_g`$ACME_est
## [1] 0.1082566
##
## `$sum_SFN_Tot Faecalibacterium diet_Alc_g`$ACME_pval
## [1] 0.164
##
## `$sum_SFN_Tot Faecalibacterium diet_Alc_g`$ADE_est
## [1] -0.06525081
##
## `$sum_SFN_Tot Faecalibacterium diet_Alc_g`$ADE_pval
## [1] 0.804
##
## `$sum_SFN_Tot Faecalibacterium diet_Alc_g`$prop_mediated_est
## [1] 2.517256
##
## `$sum_SFN_Tot Faecalibacterium diet_Alc_g`$prop_mediated_pval
## [1] 0.852
##
##
## `$sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE`
## `$sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ACME_est
## [1] 0.1387401
##
## `$sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ACME_pval
## [1] 0.116
##
## `$sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ADE_est
## [1] -0.06569426
##
## `$sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$ADE_pval
## [1] 0.696
##
## `$sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$prop_mediated_est

```

```
## [1] 1.899357
##
## `$sum_SFN_Tot f_NA_ASV516 diet_Fol.DFE_mcg_DFE`$prop_mediated_pval
## [1] 0.796
##
##
## `$sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE`
## `$sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE`$ACME_est
## [1] 0.08795385
##
## `$sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE`$ACME_pval
## [1] 0.152
##
##
## `$sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE`$ADE_est
## [1] -0.01490804
##
##
## `$sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE`$ADE_pval
## [1] 0.98
##
##
## `$sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE`$prop_mediated_est
## [1] 1.204092
##
##
## `$sum_SFN_Tot f_UCG.010_ASV667 diet_Fol.DFE_mcg_DFE`$prop_mediated_pval
## [1] 0.736
##
##
##
## `$sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE`
## `$sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE`$ACME_est
## [1] -0.1332669
##
##
## `$sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE`$ACME_pval
## [1] 0.112
##
##
## `$sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE`$ADE_est
## [1] 0.2063127
##
##
## `$sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE`$ADE_pval
## [1] 0.336
##
##
## `$sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE`$prop_mediated_est
## [1] -1.82443
##
##
## `$sum_SFN_Tot Collinsella diet_Fol.DFE_mcg_DFE`$prop_mediated_pval
## [1] 0.828
```

```
# write a for loop with the above code to make graphs of all of them
medsignif_combos <- names(med_res) # 17 of these

# make output plot list
plot_list <- list()

# for loop to make graphs
for (i in 1:length(medsignif_combos)) {
```

```

# first separate name to be three names
name <- medsignif_combos[i]
X <- strsplit(name, " ")[[1]][1]
M <- strsplit(name, " ")[[1]][2]
Y <- strsplit(name, " ")[[1]][3]

# make a filtered data frame for ggplot to use
regrplot <- med_input %>%
  dplyr::select(all_of(M), all_of(X), all_of(Y))
regrplot$trans_M <- log(regrplot[[M]]+1)
regrplot$trans_X <- log(regrplot[[X]]+1)
names(regrplot)[names(regrplot) == M] <- "M_vals"
names(regrplot)[names(regrplot) == X] <- "X_vals"
names(regrplot)[names(regrplot) == Y] <- "Y_vals"

# make graph
if(i == 1) {
  regrplot <- regrplot %>%
    dplyr::filter(regrplot$Y_vals != max(regrplot$Y_vals))
}
p <- ggplot(data = regrplot, aes(y = trans_X, x = trans_M)) +
  geom_point(aes(color = Y_vals)) +
  geom_smooth(se = TRUE, method = "lm") +
  labs(y = paste0(X, " quantity"),
       x = "ASV abundance",
       color = paste0("log transformed \n", Y, " quantity"), title = M) +
  theme(text = element_text(size = 14))

# add to plot list
plot_list[[i]] <- p
# save
ggsave(paste0("/Users/alexieva/Documents/Projects/Analysis/broccoli project/02_pubAnalysis/outputs/P1MedRes_pl", i),
       plot_list[[i]],
       height = 5, width = 7, dpi = 300)

}

g <- ggpubr::ggarrange(plotlist = plot_list,
  ncol = 5, nrow = 4,
  common.legend = T,
  legend = "bottom")

ggsave("/Users/alexieva/Documents/Projects/Analysis/broccoli project/02_pubAnalysis/outputs/P1MedRes_pl",
  g,
  height = 14, width = 18, dpi = 300)

# write a for loop with the above code to make graphs of all of them
medsignif_combos <- names(med_res) # 17 of these

# make output plot list
plot_list <- list()

```

```

# for loop to make graphs
for (i in 1:length(medsignif_combos)) {
  # first separate name to be three names
  name <- medsignif_combos[i]
  X <- strsplit(name, " ")[[1]][1]
  M <- strsplit(name, " ")[[1]][2]
  Y <- strsplit(name, " ")[[1]][3]

  # make a filtered data frame for ggplot to use
  regrplot <- med_input %>%
    dplyr::select(all_of(M), all_of(X), all_of(Y))
  regrplot$trans_M <- log(regrplot[[M]]+1)
  regrplot$trans_X <- log(regrplot[[X]]+1)
  names(regrplot)[names(regrplot) == M] <- "M_vals"
  names(regrplot)[names(regrplot) == X] <- "X_vals"
  names(regrplot)[names(regrplot) == Y] <- "Y_vals"

  # make graph
  if(i == 1) {
    regrplot <- regrplot %>%
      dplyr::filter(regrplot$Y_vals != max(regrplot$Y_vals))
  }
  p <- ggplot(data = regrplot, aes(y = trans_M, x = Y_vals)) +
    geom_point(aes(color = trans_X)) +
    geom_smooth(se = TRUE, method = "lm") +
    labs(y = "ASV abundance",
         x = paste0("log transformed \n", Y, " quantity"),
         color = paste0(X, " quantity"), title = M) +
    theme(text = element_text(size = 14))

  # add to plot list
  plot_list[[i]] <- p
  # save
  ggsave(paste0("/Users/alexieva/Documents/Projects/Analysis/broccoli project/02_pubAnalysis/outputs/P1",
                plot_list[[i]],
                height = 5, width = 7, dpi = 300))
}

g <- ggpubr::ggarrange(plotlist = plot_list,
  ncol = 5, nrow = 4,
  common.legend = T,
  legend = "bottom")

ggsave("/Users/alexieva/Documents/Projects/Analysis/broccoli project/02_pubAnalysis/outputs/P1MedRes_pl",
  g,
  height = 14, width = 18, dpi = 300)

```

Re-run Collinsella result without Vit B1 outlier person

- the significant combo from above, Collinsella - vitamin B1 - SFN, has an outlier in the vit B1 data (one individual just has extremely high B1 for some reason)

- plan is to remove this outlier and rerun just that mediation and graph

```
med_input_B1filt <- med_input %>%
  dplyr::filter(diet_Vit_B1_mg != max(diet_Vit_B1_mg))

med_input_B1filt2 <- med_input_B1filt %>%
  dplyr::mutate(quarts = ntile(diet_Vit_B1_mg, 4),
               asv_trans = log(Collinsella+1),
               sfm_trans = log(sum_SFM+1))

model.M.B1 <- lm(asv_trans ~ quarts,
                 med_input_B1filt2)
summary(model.M)
##
## Call:
## lm(formula = asv_trans ~ quarts, data = med_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0661 -0.4605  0.0160  0.3166  1.5162
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.9276     0.2604  18.922  <2e-16 ***
## quarts         0.1815     0.0968   1.875   0.0689 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6664 on 36 degrees of freedom
## Multiple R-squared:  0.089, Adjusted R-squared:  0.06369
## F-statistic: 3.517 on 1 and 36 DF, p-value: 0.06887

model.Y.B1 <- lm(sfm_trans ~ quarts + asv_trans,
                 med_input_B1filt2)
summary(model.Y)
##
## Call:
## lm(formula = sfm_trans ~ quarts + asv_trans, data = med_input)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.80901 -0.48218  0.01936  0.38929  1.30601
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.50241     0.71547   7.691 5.02e-09 ***
## quarts         0.03052     0.08422   0.362   0.719
## asv_trans     -0.03401     0.13840  -0.246   0.807
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5534 on 35 degrees of freedom
## Multiple R-squared:  0.004328, Adjusted R-squared: -0.05257
## F-statistic: 0.07606 on 2 and 35 DF, p-value: 0.9269
```

```

med_B1results <- mediate(model.M.B1, model.Y.B1,
                        treat = "quarts",
                        mediator = "asv_trans",
                        control.value = 1, treat.value = 4, # lowest and highest quartiles
                        boot = TRUE, sims = 500)

modsumB1 <- summary(med_B1results)
modsumB1
##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##           Estimate 95% CI Lower 95% CI Upper p-value
## ACME           -0.232      -0.626       0.01  0.064 .
## ADE             0.451      -0.181       1.02  0.160
## Total Effect    0.219      -0.499       0.84  0.548
## Prop. Mediated  -1.057      -6.299      17.55  0.596
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Sample Size Used: 37
##
##
## Simulations: 500

coeff_tab_medB1 <- list(ACME_est = modsumB1$d.avg,
                      ACME_pval = p.adjust(modsumB1$d.avg.p, method = "fdr"),
                      ADE_est = modsumB1$z.avg,
                      ADE_pval = p.adjust(modsumB1$z.avg.p, method = "fdr"),
                      prop_mediated_est = modsumB1$n.avg,
                      prop_mediated_pval = p.adjust(modsumB1$n.avg.p, method = "fdr"))

```

save image chunk

```

save.image("/Users/alexieva/Documents/Projects/Analysis/broccoli project/02_pubAnalysis/Prediction1_med

```